

Rice — Crop Production Notes

Introduction and Local Names

The scientific name of rice is *Oryza sativa* L., and it belongs to the family Poaceae. In Urdu, rice is called Dhan or Chaval, and in Punjabi it is known as Munji.

Crop Characteristics

Rice is an annual crop with hypogeal germination. It is a short-day, Kharif-season crop with a determinate growth habit and is self-pollinated. Its seed is referred to as grain.

Origin

Rice is believed to have wild forms in Asia, Africa, and America. *Oryza sativa* (Asian rice) contains two major subspecies: the sticky, short-grained japonica (or sinica) variety, and the nonsticky, long-grained indica variety. *Oryza glaberrima* is the African species. Rice cultivation is believed to have begun about 5,000 years ago in China, and it reached the USA in the 17th century.

Top Rice Producing Countries

According to FAO (2013) data, the leading rice-producing countries and their production figures (in tonnes) were as follows: China produced 203,612,192 tonnes, India produced 159,200,000 tonnes, Indonesia produced 71,279,712 tonnes, Bangladesh produced 51,500,000 tonnes, and Vietnam produced 44,040,457 tonnes.

Rice Production in Pakistan (2014)

In 2014, the area under rice cultivation in Pakistan was 2,890,646 hectares, production was 7,002,832 tonnes, and the yield was 2.42 tonnes per hectare. In 2014, Pakistan ranked 13th in world rice production.

Province-wise Contribution (2010–13)

Of the total varieties grown, 25% were hybrid and 75% were conventional varieties.

By area under cultivation: Punjab accounted for 71% (1.8 million hectares), Sindh 22% (0.75 million hectares), Baluchistan 5% (0.18 million hectares), and KPK 2% (0.06 million hectares).

By production: Punjab accounted for 63% (3.4 million tonnes), Sindh 31% (2.8 million tonnes), Baluchistan 4% (0.6 million tonnes), and KPK 2% (0.1 million tonnes).

Importance

Rice is the second major cereal after wheat. It provides about 60% of the energy needs of half of the world's population. Rice occupies about 10% of the cropped area in Pakistan, including the Kallar

tract. The grain contains about 80% carbohydrates and 7% protein. Pakistan ranks 4th in the world in rice exports.

Climate Requirements

Rice grows well in hot, humid tropical conditions, requiring about 1000 mm of annual rainfall and a temperature of around 20°C.

Soil Requirements

Rice grows best in soils with 40–60% clay content. It is not successful in sandy soils but is fairly resistant to salinity.

Growth Stages of Rice

- Germination
- Seedling growth
- Tillering
- Stem elongation
- Booting
- Inflorescence
- Anthesis
- Milk development
- Dough development
- Ripening

Root Development

Rice roots are fibrous and shallow. They contain aerenchyma cells, which supply oxygen to the plant while it stands in water.

Shoot Development

The rice shoot has a jointed stem with hollow internodes and pithy nodes. Coarse varieties are short-statured, while fine varieties are long-statured.

Inflorescence

The rice inflorescence is a branched spike, with each branch ending in a grain. The grain has an outer covering called the husk or hull. When the husk is not removed, the grain is called paddy (unpolished); when the husk is removed, it is called rice (polished).

Shelling

Shelling refers to the removal of the husk. Paddy can germinate, but rice (after husk removal) cannot. In Pakistan, Basmati rice undergoes a hot water (heat) treatment before shelling, which results in less breakage and easier cooking.

Varieties — Grain Type

Coarse varieties have rough leaves and are short-statured, while fine varieties have plain leaves and are long-statured.

Nursery

A nursery of 6–8 marlas is sufficient to transplant one acre. Transplanting is done after puddling. Nursery raising methods include the dry method, wet method, Rab method, and seed priming. A nursery that is 25–30 days old can be transplanted.

Wet method: This is practiced in clayey and clay loam soils where standing water can be maintained. The selected field is filled with water 25–30 days before sowing the nursery. It is better to plow the field twice under dry or watery conditions to break soil clods and create better puddled conditions. After watering, the field is plowed twice, followed by planking; this is repeated after 7–10 days depending on water availability.

Dry method: This is practiced under dry soil conditions. Seed beds of convenient dimensions are prepared by raising the soil to a height of about 5–10 cm. A thin layer of farmyard manure or half-burnt paddy husk may be spread on the nursery bed to facilitate uprooting. Soaked seed is spread over the seedbed and then irrigated.

Rab method: The seed bed is covered with 3–5 inches of dry refuse, which is set on fire in April–May. This results in some loss of nitrogen but faster availability of other nutrients such as potassium, which is usually the limiting factor.

Seed priming: Seeds are primed using CaCl_2 and KCl salt solutions, a practice used by farmers in Sialkot and Faisalabad districts. Nursery seedlings raised this way are healthy and vigorous, and are ready for transplantation within 25–30 days. Seedlings from primed nurseries show better growth, yield better kernel quality at harvest, and show greater disease resistance.

Direct Sowing

Direct sowing involves irrigation, seed bed preparation at the proper moisture condition, and drilling of soaked or treated seeds.

Transplanting

Transplanting involves flooding (heavy irrigation), puddling, and then transplanting the seedlings.

Sowing Time

- Fine varieties: 1st June to 1st July
- Coarse varieties: 20th May to 1st June
- Transplanting is done at 25–30 days of nursery age
- Direct sowing is done earlier than transplanting
- Hill distance: 25 cm

Direct sowing is generally carried out around the time of nursery sowing, or about two weeks after the stated sowing time. A "hill" refers to the row-to-row and plant-to-plant distance point where seedlings are transplanted. Two seedlings are placed per hill for a 25–30 day old nursery, and four seedlings per hill for a nursery older than 40 days.

Fertilizer

- Coarse varieties: 200:75:60 kg/ha (NPK)
- Fine varieties: 125:60:60 kg/ha (NPK)

Irrigation

Rice requires 15–16 irrigations, with standing water maintained throughout and no dry period. Weedicides are also used as part of water management.

Weeds

Weed management in rice relies on standing water, along with cultural and chemical control methods.

Weed Management

Weeds can cause 15–20% yield losses, and sometimes up to 50%. There are three groups of weeds found in rice fields: weeds of the grass family, weeds of the sedge family, and broad leaf weeds.

Common Rice Weeds

Weeds of the grass family:

- Swanky grass (*Echinochloa colona*)
- Dhedan (*Echinochloa crusgalli*)
- Khabal grass (*Cynodon dactylon*)
- Narro (*Paspalum distichum*)

Broad leaf weeds:

- Mirch boti (*Sphenoclea zeylanica*)
- Chopti (*Marsilea minuta*)
- Darai boti (*Eclipta prostrata*)

Weeds of the sedge family:

- Ghoein (*Cyperus difformis*)
- Bhoein (*Cyperus iria*)
- Deela (*Cyperus rotundus*)

Insect Pests

Major insect pests of rice include the Pink Rice Borer and Yellow Stem Borer (both stem borers), and the Leaf Hopper, which causes the whole field to appear white when infestation is heavy.

Diseases

Common rice diseases include Root Rot, Collar Rot, and Stem Rot. Rice Blast (locally known as Jhulsao) can cause up to 80% yield reduction.

Harvesting

Early maturing varieties are harvested 90 days after transplanting, while late maturing varieties are harvested 140 days after transplanting. Harvesting time is indicated when the panicle turns golden. Coarse varieties mature earlier than fine varieties.

Rice Varieties by Province

Punjab — Fine: Super Basmati, Basmati 2000, Basmati Pak, Basmati 370, Basmati 198, Shaheen Basmati, Basmati 515.

Punjab — Coarse: IRRI-6, NIAB IRRI-9, KSK-133.

Sindh — Fine: Shadab, Khusboo 95, Shua 92, Sada Hayat, Latifi, DR-82, DR-83.

Sindh — Coarse: IRRI-6, IRRI-8, KS-282.

KPK — Coarse: IRRI-6, Fakhr-e-Malakand, KS-282.

KPK — Fine: JP-5, Basmati 385, Swat 1, Swat 2, Pakhal.

Baluchistan — Fine: Basmati 385, DR-83, Sada Hayat.

Baluchistan — Coarse: IRRI-6, KS-282.